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RESEARCH ARTICLE

Designing automated digital safety systems for building health monitoring on construction sites

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ABSTRACT Uncertainties in construction projects clarify a central problem in risk assessment and safety control of the background building construction during their lifecycles. Digital safety systems are presented for building an operated diffusion center for gathering information and possibly developing a digital twin in the safety construction management process. The essential technologies were reviewed and investigated to select proper systems that aim to automate the safety process with good information gathering. First, in this study, the role of digital-system elements is examined in shaping a diffusion center for gathering information and seeking to model integrated safety information. Then, it offers the data-physics-driven model for automating digital safety systems processes in minitoring construction site health and safety. A multi-criteria decision analysis is used to sort the alternatives. The results demonstrate that the selected technologies are required to gather enough information to shape a diffusion center for automating the safety process in construction management and building health monitoring. The results show that the model assumption has been validated by the value of "1.684" as the table value of T 95% confidence interval for all components. Additionally, an investigation into the Tehran municipality region demonstrated the applicability and usefulness of the method in practical settings. Furthermore, the selected digital safety strategy in collaboration with SEI/ASCE 7-02 has been proven to be implementable for all types of buildings mentioned in this study. The findings demonstrate that the methodology employed in this study can be used as a reliable tool for ensuring the safety of buildings. This study also identifies the pre-construction period (PrCP) as a primitive emergency for creating safety values for all types of buildings. The estimated values for very high importance, high importance, and medium importance buildings are in order 34%, 35%, and 31%, respectively, and also indicating that the PrCP can be the most safety data-driven stage of construction.

KEYWORDS safety management system, construction safety management, digitalized systems, technique for order of preference by similarity to ideal solution, decision making

1 Introduction

The digital side of construction indicates efficiency in managing predictable challenges, and it leads to being effective in discovering uncertainties of projects. In this avenue, data interactions and relative simulation models benefit from driving values in a project, and the future of each project will be independent of this process that contains a data-driven and value-created structure to shape real-time communication for the achievement of best practices in construction management. Due to this fact, all information related to projects is communicated in cyberspace in the form of digital by design. This situation is an opportunity to optimize the project management process of threats and risks ahead of restraint.

The construction industry faces significant digital risks due to the complex nature of embedding safety monitoring construction equipment in the digital

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environment [1]. The Global Construction Rate Trend Report highlights the rapid evolution of cyber threats in the construction industry [2,3]. Additionally, the complex construction environment and environmental issues contribute to construction risk factors [4]. The occupational fatality and injury rates in building construction are relatively high, particularly in dynamic construction work environments [5,6].

To address these challenges, several researchers have proposed solutions such as the development of a standard vulnerability scoring system [7], spired security systems for smart buildings [8], addressing cyber-security risks in intelligent buildings [9], and exploring the potential use of blockchain technology in construction management [10]. However, there is a need to integrate these various safety methods, systems, tools, and technologies into a cohesive digital safety system (DSS).

Construction management requires practical techniques to ensure safety and effectiveness throughout the process [11]. Safety plays a critical role in buildings construction management (BCM) [12,13]. However, current digital safety tools lack integration and coherence, which leads to limited effectiveness [14]. Therefore, there is a need to merge different safety methods, systems, tools, and technologies into an activated DSS.

This study proposes a comprehensive framework that covers the pre-construction, construction, and post-construction periods of the building construction process. By integrating safety trends and application systems, and this framework aims to create an activated DSS that enhances safety management throughout the construction lifecycle. While digital tools have been applied in construction safety management, limited research has been done to conceptualize the role of digital technologies in safety management and accident prevention [15–17]. This research aims to demonstrate the importance of shifting safety trends in the construction industry and proposes an automated DSS as a solution. By examining the relationship between digital technologies and safety performance [18], this study aims to explore research novelty in construction safety management. It also highlights the need for collaboration with stakeholders to ensure the successful adoption of new safety processes and the potential for further advancements in construction safety through continued research and development.

2 A comprehensive literature review

The literature review section provides an overview of digital safety in construction management, emphasizing the integrated approach of digital systems in building construction processes [19]. It explores the role of digital tools in the design, construction, and maintenance phases, discussing their importance in enhancing safety management.

Digital safety in construction management refers to the integrated utilization of digital systems to enhance safety throughout building construction processes [19]. This approach involves transforming operational tasks and processes into a digital environment, aiming to minimize failures and promote a safer construction environment. The automation of construction operations through digital transformation diffusion centers plays a crucial role in streamlining the construction safety process [19,20]. The process includes design process, construction, and maintenance, which are explained in the following sections.

2.1 Design process

The design process, as defined by the American Institute of Architects, encompasses schematic design, design development, contract documents, bidding, contract administration, and task assignments [21,22]. This essential part of the design process deals with a wide range of information items that can be linked to building information modeling (BIM). Digital strategies are increasingly recognized as an essential aspect of the design process, and several definitions of the design process highlight the application of digital tools and detailed design stage variables [23,24]. Furthermore, the integration of digital context solutions in the design stage of building construction has been instrumental in minimizing environmental dangers [25,26]. The early stage of the design process is critical for incorporating digital context and considering safety factors to prevent accidents or incidents [18,27]. Therefore, the measurement of safety factors using digitally equipped tools is essential for effective safety management in building construction.

2.2 Construction

The building construction industry is a complex system characterized by multiple interrelated production processes, prioritized decision-making, operational cost challenges, quality assurance requirements, inherent uniqueness, and safety concerns [28–33]. The advent of intelligent construction, which involves the interaction between humans and intelligent machines, has further compounded the challenges posed by volatile construction conditions [34]. The presence of multiple systems and the interaction between humans on construction sites can potentially lead to hazardous occurrences. To address this complexity, exemplary safety monitoring with advanced digital tools is crucial. Digital safety tools play a vital role in detecting and responding to hazards, thereby reducing risk exposure within the construction environment.

2.3 Maintenance

Constructed buildings face various risks, such as earthquakes, fire, overheating, collapses, floods, and interconnected risk relations [35–40]. Risk occurrences can result in crises with disastrous consequences. Recognizing the importance of building safety, the Building Safety Act 2022, proposed by the UK government, empowers homeowners with increased authority to make homes across the country safer [41]. The act focuses on the maintenance stage of building life, promoting sustainability, and implementing risk-control-based practices to prevent hazards. The maintenance process in buildings has become increasingly challenging due to the complexity of building systems [42–44]. Monitoring higher-risk facilities to control hazardous situations is crucial, leading to the development of integrated platforms for real-time risk monitoring and threat detection [45–48]. The operation and maintenance phase of the building life cycle significantly impacts the environment [49], and studies have indicated that the overall cost of a building's life cycle is primarily incurred during the maintenance process [50]. Proper safety monitoring, supported by an effective and efficient functional framework, is necessary for maintaining a building's environment and improving productivity while mitigating rising costs associated with inefficient maintenance practices [51].

As a technical approach, the computational intelligence methods [52–54] has the potential to solve complex problems with advanced numerical methods and applied mathematics. The below approach addresses energy-safety challenges in the building environment and their intelligent solution examples.

2.4 Energy-safety and deep energy methods

Energy consumption in buildings has remained a public safety issue in urban areas. This problem is dedicated to an energy-safety approach. In response to the energy-safety challenges, using machine learning (ML) hybrid models, advanced predictive models, as deep energy methods, are considered to maximize the function of energy systems [55]. These challenges are addressed due to the lack of efficiency and sustainability of techniques in handling big data in energy systems, and in other cases as the structural energy element problem. The hybrid deep learning techniques have indicated a good performance in optimizing renewable energy systems [56].

Intelligent computational methods are essential solutions to energy-safety challenges. They enhance buildings' energy efficiency, safety, and durability by providing advanced tools for modeling, predicting, and optimizing structural and environmental performance. The below describes the advanced ML methods for physics-informed modeling and simulation in engineering mechanics.

1) Adaptive Collocation Method. This research [57] introduces an adaptive collocation method for solving second-order boundary value problems, such as Poisson's and Helmholtz equations. It dynamically selects collocation points based on residual values from previous training steps, enhancing robustness in non-smooth regions through strategic point density adjustments.

2) Parametric Deep Energy Method (P-DEM). A P-DEM is proposed for elasticity problems [58]. It incorporates strain gradient effects and leverages physics-informed neural networks (PINNs) to minimize a potential energy-based cost function. This method achieves faster convergence and simplified implementation using NURBS basis functions and Gauss quadrature.

3) Physics-Informed Neural Network for Brittle Fracture. A novel PINN algorithm is developed for brittle fracture problems [59]. It minimizes the system's variational energy rather than the residual of governing partial differential equations, ensuring exact boundary conditions, faster training, and superior accuracy over conventional residual-based PINNs through transfer learning.

4) Deep Autoencoder-based Energy Method (DAEM). The DAEM is introduced for analyzing bending, vibration, and buckling in Kirchhoff plates, combining a deep autoencoder with the minimum total potential principle for unsupervised feature learning, implemented using PyTorch and the LBFGS optimizer with transfer learning to improve computational efficiency and generality [60].

3 Methodology

This research aims to shape an automated DSS for building health monitoring in construction sites. By leveraging data-driven approaches and integrating various safety methods, systems, tools, and technologies, this system aims to improve safety, reduce risks, and enhance construction site health monitoring. The subsequent sections of this study are divided into the research methodology, results, and recommendations for organizations responsible for safety in construction buildings. Furthermore, other aims lead to cover a range of functional selection systems that can be applied in the automated digital systems for health monitoring in building construction by focusing on the three key phases of the construction process.

In the digital age, safety management plays a crucial role in ensuring a safe construction period (CP) for building projects. With the use of advanced technologies such as BIM and the Internet of Things (IoT), safety

managers can monitor potential hazards in real time, analyze data to identify patterns, and take proactive measures to mitigate risks, ultimately ensuring a safer construction environment for workers and the community. This research novelty focuses on the following.

1) Representing the evolution of digitalization and automation in construction safety management.

2) Introducing novel research approaches to improve safety in construction buildings.

3) Identifying gaps in current safety measures and proposing new solutions.

4) Addressing challenges in implementing innovative safety technologies.

5) Examining the impact of novel safety measures on construction productivity and efficiency.

6) Collaborating with stakeholders to ensure the successful adoption of new safety processes.

7) Analyzing the potential for further advancements in construction safety through continued research and development.

Overall, this research aims to contribute to the advancement of safety management in the construction industry by exploring the potential of digital technologies, proposing an automated DSS, and providing valuable insights and recommendations for organizations responsible for safety in construction buildings.

This study was done using three steps: one sample, drawing normally distributed curves of data, and a Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS), as shown in Fig. 1. The method involves a systematic approach to evaluating the performance of a set of alternatives. First, the one-sample technique was used to test whether the mean value of a particular variable is significantly different from a known value. Second, normal distribution curves were drawn to visualize the data and gain insights into its distribution characteristics. Finally, the TOPSIS method was applied to rank the alternatives based on their relative closeness to an ideal solution and select the best one. Overall, this methodology provides a rigorous framework for decision-making in safety management domains, such as control actions, sharing safety information, and building health monitoring.

The one-sample *t*-test is a statistical method that compares the mean of a sample to a known or hypothesized population mean. Its main aim is to determine this parameter if there is a statistically significant difference between the two means. This test involves several stages, including stating hypotheses, calculating the *t*-statistic, and comparing it to a critical value to find out if there is a statistically significant

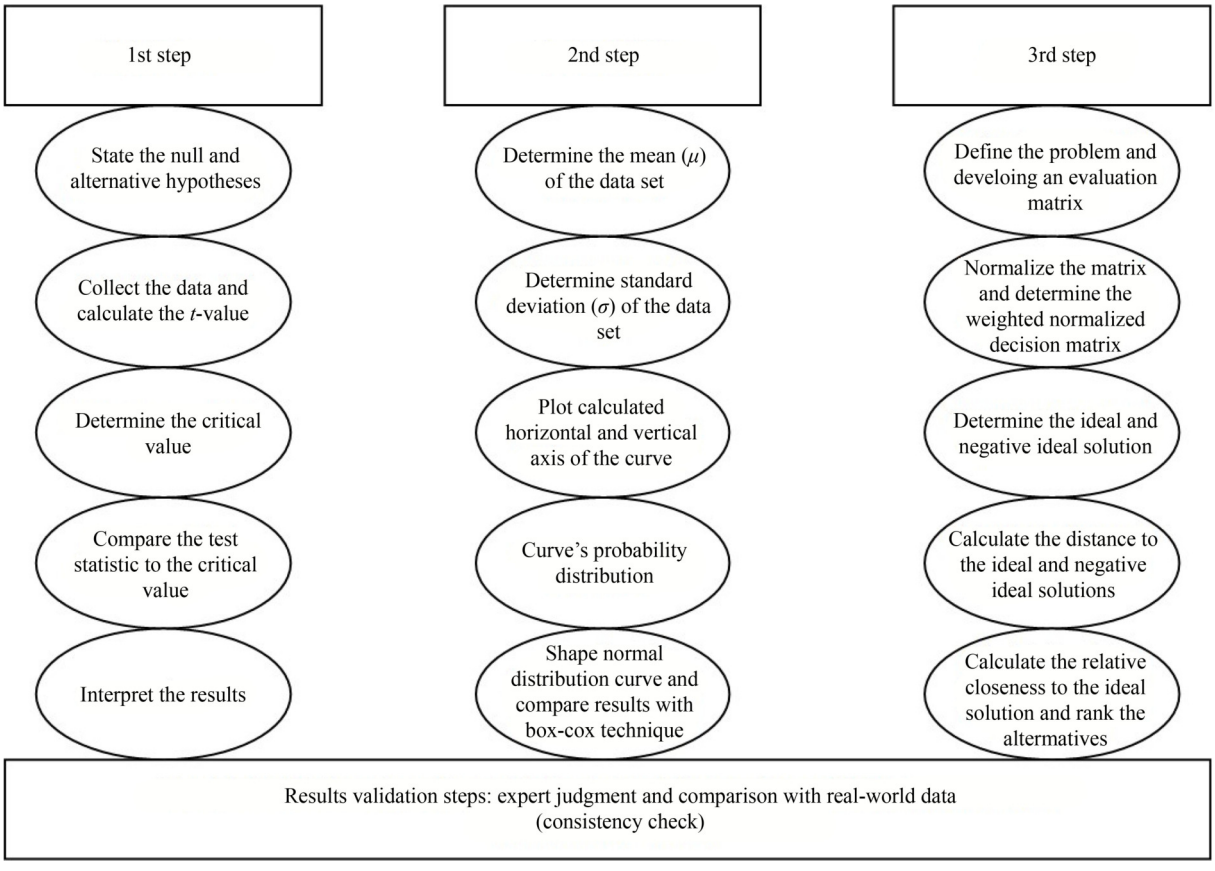


Fig. 1 The direction of the methodology's mind map.

difference. The stage of this method is described in the first step, as shown in Fig. 1, in which the hypothesis test of the study is mentioned.

In the next step, the probability distribution of a random variable with normal distribution curves is clarified. For shaping related curves, after determining the mean (μ) and standard deviation (σ) of the data set, it is required to use the below formula.

$$f(x) = \left(\frac{1}{\delta\sqrt{2\pi}}\right) \times e^{\left(\frac{-(x-\mu)^2}{2\delta^2}\right)}, \tag{1}$$

where $f(x)$ is the height of the curve at a given value of x. Then, values achievement was plotted to shape curves. It is dedicated to the research methodology because it predicts the likelihood of certain values occurring within a given data set and quantifies the variability of the data around the mean. Therefore, the results help rank options for decision-makers and provide more transparency to the data.

Finally, TOPSIS is used to identify the best solution from a set of alternatives based on their similarity to an ideal solution and dissimilarity to a negative solution.

TOPSIS is a method that uses Euclidean distance to determine the relative proximity of the options to the optimal solution, compare the options, and finally select one of them [61]. In this approach, the positive ideal solution defines the best values for each achievement attribute, while the negative one indicates worse values. This method can also demonstrate the distance of ideal solutions to both positive and negative solutions in order to clearly define a decision based on selected criteria. For ranking options, the normalized values must be calculated using the TOPSIS method and then multiplied by weights, which were determined by experts. Entropy is a method that is frequently used in determining attribute weights for TOPSIS [62]. Using the below equations, worst and ideal solutions are determined. This part is evaluated by Smart TOPSIS software.

$$A_w = \{\langle \max(v_{ij} \mid i = 1,2,\ldots,m) \mid j \in J_- \rangle, \langle \langle \min(v_{ij} \mid i = 1,2,\ldots,m) \mid j \in J_+ \rangle \rangle\} \equiv \left\{v_j^+ \mid j = 1,2,\ldots,n\right\}, \tag{2}$$

$$A_b = \{\langle \min(v_{ij} \mid i = 1,2,\ldots,m) \mid j \in J_- \rangle, \langle \langle \max(v_{ij} \mid i = 1,2,\ldots,m) \mid j \in J_+ \rangle \rangle\} \equiv \left\{v_j^- \mid j = 1,2,\ldots,n\right\}, \tag{3}$$

where J_+ and J_- are associated with the criteria having positive and negative impacts, respectively. v_{ij} is defined as the value of indicator i for alternative j. v^+_j and v^-_j are defined as the best and worst alternatives with respect to indicator i. A_w and A_b are defined as the ideal and worst values with respect to indicator i.

Using Eqs. (4) and (5), the distance from the ideal solution and the worst solution can be measured, respectively.

$$D_i^+ = \sqrt{\sum_{j=1}^{n}(v_{ij} - v_j^+)^2},\ i = 1,2,\ldots,m, \tag{4}$$

$$D_i^- = \sqrt{\sum_{j=1}^{n}(v_{ij} - v_j^-)^2},\ i = 1,2,\ldots,m. \tag{5}$$

Using the following equation, proximity to the ideal solutions can be determined, and the options can be sorted.

$$C_i = \frac{D_i^-}{(D_i^+ + D_i^-)},\ i = 1,2,3,\ldots,m. \tag{6}$$

By the use of TOPSIS, decision-makers evaluate and analyze a set of alternatives with multiple attributes to provide desirable alternatives for supporting decision makings. This approach is designed to respond to complex problems, and it is a part of modern decision science [63].

3.1 Theoretical model

The leading causes of construction injury are lack of integration in safety systems, insufficient safety design features, or both. Lack of integration systems in safety may arise through insufficient rights of cultural governance, inadequately maintained equipment, lack of communication between designer and construction management, inadequate supervision, or a combination of these factors. Insufficient safety design features mainly arise through inadequate awareness about the application of merges systems, lack of application in technical tools, etc. given the importance of the digital safety model during the construction periods, it is not surprising that safety improvements focused on digital strategies have acquired popularity in the development of safety performance. However, studies on digital safety for construction management are somewhat limited. There are several reasons for this; first, construction is a hazardous occupation, and it was initially believed that risk management (RM) strategies alone would result in better safety performance. Second, the construction industry is practically regulated.

3.2 Variables

3.2.1 Safety integrated model in pre-construction period (PrCP)

During the safety-integrated model in the PrCP, the implementation of a safety-integrated model brings significant benefits. This approach involves incorporating

safety considerations into the BIM model from the early stages of project development. By doing so, architects and designers can accurately assess and analyze potential safety risks before construction begins. This proactive approach allows for the identification of hazards, clashes, or shortcomings in the design, enabling necessary adjustments to be made to prevent accidents and improve overall safety standards. The Safety Integrated Model in the PrCP acts as a valuable tool for RM, enhancing collaboration among stakeholders and facilitating informed decision-making that prioritizes safety throughout the entire construction lifecycle.

1) Building Information Modeling

There are many high potentials for BIM usage in a digital environment to reduce and control predictable risks. Safety integrated BIM model helps architects and structural designers to achieve their alternatives by calculating construction safety risks. Consequently, Autodesk Revit's digital environment has developed by linking a plug-in with safety risk data [27]. In this way, the ultimate power of integration for complex projects is revealed. Fails to detect malfunctions specify to prevent the faults in real construction time. Therefore, many design process gaps are uncovered to prevent potential risk occurrences. Management issues for safety may be used to indicate the level of risk in the projects; an integrated approach to the application of BIM is helpful for safety improvement in this area [64].

• H1: BIM application is practical in the designing process's PrCP.

2) Geographic Information System (GIS)

The GIS provides a holistic approach to enhance construction safety by integrating detailed environmental information. Utilizing GIS for safety planning considers multiple environmental factors such as site conditions, topography, thermal comfort, and access route planning. These factors play a crucial role in directly influencing the safety of workers on-site.

GIS provides an approach to considering construction safety from the macro perspective as it contains detailed environmental information. During the safety review process, if a planned sequence results in a hazardous situation, it may be corrected within GIS itself before actual implementation. Additionally, 4D modeling, topographical conditions, and a safety database in a single environment assist safety planners in examining what safety measures are required, when, where, and why [65]. This fact helps engineers make the right decision regarding harmful risks.

GIS is a computer-based technology that uses GIS to manage and combine data, solve problems, or understand past, present, and future situations. In this way, it can map geographical features and relationships between these features on the ground, which is the best option for placing the map on the construction site. The real-time detection function reduces the main risks in construction sites, such as floods, earthquakes, wind, etc.

• H2: GIS leads to finding the best position for buildings and provides safe locations in good condition.

3) Risk Identification and Assessment (RIA)

RIA is an essential process that the whole safety and quality standards emphasize instructions to implement for risk detection [66–68]. To achieve a safe environment, the method of RM must be implemented to prevent the hazard [69]. There are numerous directives, standards, regulations, and regulatory guides that establish the safety authority to develop and maintain construction site conditions as safe. Generally, codes and standards denote securing safety by following essential rules and practices. These codes and standards play a significant role in ensuring safety by defining essential rules and practices [19]. At the moment, there are several such standards to direct the way of risk detection step by step, as introduced in the initial process of the paragraph.

• H3: Risk assessment and evaluation of the status are the factors that will be affected during the PrCP.

3.2.2 Safety integrated model in construction period

The safety integrated model in the CP refers to a comprehensive approach that encompasses various safety measures and practices to ensure a secure working environment throughout the construction process. This model involves the integration of safety guidelines, RM strategies, and safety protocols into the planning, execution, and monitoring phases of construction projects. By incorporating safety considerations from the early stages, such as design and site layout, to the final stages of constructing a building on site, including equipment operation and worker training, and etc. the safety integrated model in CP aims to prevent accidents, mitigate risks, and enhance overall safety standards on construction sites.

1) Risk Management

Efficient risk management RM plays a crucial role in alleviating project-related tensions and ensuring smooth process execution. RM relies on the measurement of predictable values. Nonetheless, uncertainties are always a threat to the project environment. So, it is not just RM for implementation to control hazardous conditions of the projects. It is placed in three factors that are connected firmly to the construction industry's safety performance [70]. In Ghana, project managers of enormous construction firms require training to recognize systematic RM practices for achieving value for money in a construction project [71]. Similarly, in Iran's construction processes, where financial resources are prioritized, employing RM techniques becomes essential. By doing so, building construction projects can optimize value and cost, which are critical considerations aligned

with the project management body of knowledge standard [66].

• H4: RM methods are used in construction safety.

2) Safety management system (SMS)

Inadequate management skills often result in an unsafe project environment with insufficient safety measures. To address this, a SMS offers a systematic approach to controlling hazards by integrating them into a logical sequence of activities. Implementing a practical and proactive SMS on construction sites enables the real-time detection of unpredictable hazardous events [72]. The SMS can be structured to identify failures within a defined system or programmed to overcome challenges encountered during construction projects. Therefore, it is highly recommended that an efficient and effective SMS be designed specifically tailored to building construction projects and their stakeholders. Unfortunately, there is currently no standardized tool available for auditing the safety processes of building construction and building construction companies need it. Therefore, these companies should provide safety incentives and actively engage in developing the designed SMS [73].

• H5: Construction safety is followed by an integrative SMS.

3) Legislation Governance (LG)

Rules and regulations serve as guiding principles for stakeholders to follow in a proper manner. However, some construction executives choose to disregard these established principles, believing that it will help them minimize costs. Despite such challenges, legislative governance plays a crucial role in ensuring that safety remains a priority throughout the lifespan of a construction project. By adhering to safety regulations, all aspects of the building, including its functions, are guaranteed to meet the expectations of customers. Safety is also closely linked to cost management and overall well-being. Given the importance of safety, there is a need for numerous directives, standards, regulations, and regulatory guides to establish clear principles that promote safety objectives within LG, which should play an active role in monitoring the implementation of these step-by-step control methods. LG is an essential part of implementing the rule of law in construction safety as the standard.

• H6: Safety instructions are followed at LG

3.2.3 Safety integrated model in the post-construction period (PCP)

The safety integrated model in the PCP refers to a comprehensive approach aimed at maintaining safety and mitigating risks even after the completion of a construction project. This model recognizes that safety considerations should extend beyond the construction phase and continue throughout the lifecycle of the building. It involves implementing measures such as regular inspections, maintenance protocols, and adherence to safety standards to ensure ongoing safety and prevent potential hazards. By incorporating safety practices into the PCP, the safety-integrated model aims to create a safe environment for occupants, address any potential issues that may arise over time, and ensure the long-term durability and functionality of constructed buildings.

1) Sensing and Warning Systems (SWS)

Sensors play a crucial role in detecting hazardous events and dangerous occurrences in a timely manner, making them essential for implementing preventive actions and warning systems. These technologies have the potential to significantly reduce safety risks in the PCP at an operational level. Location sensing techniques like radio frequency identification [74] and ultra-wideband nodes [75] can be applied as security systems in buildings to enhance safety through improved sensing and warning capabilities. These technologies enable the creation of active warning systems that protect the built environment from risky situations during the ongoing maintenance of buildings. They can also be utilized to enhance existing safety recognition systems in buildings, such as fire protection, air conditioning, and smoke evacuation systems, collectively referred to as SWS.

SWS relies on technology with real-time hazard detection capabilities and the ability to respond promptly to crises. Furthermore, these systems can be designed to align with the safety objectives of buildings by establishing logical interactions between conditions and responses.

• H7: Sensors and warning systems are in use as protective equipment at maintenance process of buildings (M.P.B).

2) Internet Environment Services (IES)

The Internet is a worldwide broadcasting capability, a mechanism for information dissemination, and a medium for collaboration and interaction between individuals and their computers without regard for geographic location [76]. The Internet is an open environment world with virtual networks, which have almost all the infrastructure needed to use a digital soft environment for safety. For this reason, this virtual world, the Internet, can store and transfer safety information flow analysis for safety experts or safety managers who need to make their decisions correctly. At the same time, they find the condition of the building to be risky, or they can take valuable feedback about safety information in a crisis to handle it. Another thing is that they shape an approach to the evolution toward integration in date-based management. Applying IES in PCP is safety information mining, which lets the safety managers make meticulous decisions on safety directionless of building care. Also, there are more ways to use this virtual world that lets the safety manager choose updated safety information about what will be the base of making a decision suitable.

• H8: Virtual Environments and internet websites are the most effective basements for evaluating the safety and safety culture.

3) Virtual Reality (VR) and Virtual Prototype (VP)

VR is a technology that uses computers, software, and peripheral hardware to generate a simulated environment for its user [77]. Applying these technologies in building construction can help to increase the reliability of investors and construction safety training. This research [78] strongly recommended incorporating VR in construction safety training, given the need for improved training and the advantages of using VR. When the building is constructed and placed in the maintenance process, the stakeholders need to present its characteristics to customers as a lifetime product with a high-level price. The best way to introduce the product to customers is to use VR and VP technologies, guaranteeing a reliable sale for stakeholders. This issue means it is like a bridge to success to bring money back or investment back to investors. Another technology as the same application is VP, which is a computer-aided design process concerned with the construction of digital product models ("VPs") and realistic graphical simulations that address the broad issues of physical layout, operational concepts, functional specifications, and dynamics analysis under various operating environments [79] [80]. Both mentioned technologies enable the PCP Sales guarantee and provide the platform for practitioners to try before building' the project.

• H9: VR Technology is effective for safety training in the useful life of buildings.

4) Software's Supported Risk and Safety Controls (SSRSC)

To reduce human errors, it is better to monitor safety indicators needed to implement the computer-based software that can calculate input dates with a logical mathematics model for safety condition reports, like a user interface between safety engineers or safety managers with items of information. Rouse [81] believed that introducing computers would improve safety, which is true about the PCP. In this approach, another primary strategy to achieve the safety aims of PCP is to dedicate at least one SSRSC system for monitoring risky conditions. This is because digital computers have the potential to provide increased versatility and power, improve performance, greater efficiency, and decrease cost [82], especially in safety role implementation.

• H10: Safety and risk management software are used to monitor the safety rate at M.P.B.

5) Safety Audit System (SAS)

This mechanism delivers safety aims to a proactive mode of control, which brings up strengths with opportunities and brings down weaknesses with threats. It helps reduce the risk impact for PCP, but it is done by a safety manager responsible for handling risky conditions with risk reduction strategies. A SAS is based upon the periodic review of the system safety process, and safety auditing is a systematic method to evaluate a company's SMS [83]. It also must be implemented by independent safety groups like enterprises or private companies. Consequently, the introduced mechanism will impact PCP if they connect to SSRSC. An outcome such as this structure will allow the operators to find faults and malfunctions in the system of dynamic maintenance of buildings in real time.

• H11: All safety practices are active in the M.P.B.

4 Data and analysis

4.1 Sample

The setting for this study was Tehran Municipality Region One (TMR1) in Iran. This environment was selected because there are many construction management and safety experts, and the possibility of responding to the questionnaire forms a lot of work experience. Also, the selected region is a centric placement in Tehran. It is specialized with high habitability needed to handle safety. The TMR1 is comprised of more than 50 municipal employees who are responsible for positions such as safety experts, construction managers, civil engineers, etc. The sample consisted of 49 municipal employees who participated in this study (response rate 90.75%).

Among the participants, 86% were recorded as related to construction expertise, and 14% of them were safety experts who had experience in construction projects. Furthermore, Among the participants, 19% of them had 16–20 years of work experience, 31% of participants had between 5 and 11 years of work experience, 20% had more than 20 years of work experience, as well as 30% of them had less than ten years of work experience. The experts were characterized by their work experiences, expertise areas, and educational qualifications. They were engaged with construction projects in the Tehran region, and they believed the mentioned demonstrated models of this research should be applied to all projects for increasing productivity in their construction projects. They believed in the safety trends of the construction sites. The research was essential to find who was confronted with these areas of work experience. Regarding educational qualification, of the participants, 12% had a doctor of Philosophy, 33% a master of Science, and the rest had a Bachelor of Civil Engineering or Bachelor's Science.

4.2 Instruments

Measures used in this research were GIS, BIM, RIA, RM,

SMS, LG, SWS, IES, VR, VP, SSRSC, and SAS. All cases are categorized into pre-construction, construction, and post-construction periods, as shown in the theoretical section. All the required information was collected through a questionnaire. The questionnaire consisted of three parts, each about one of the above components. The scale items mean and standard deviation of each of the three constructs are shown in Fig. 2, which is divided into three groups. The construction supervisors, managers, and engineers obtained each respondent's digital safety forms. These were asked to evaluate the criteria used in this research—the eatery items of information given to them to direct the proper selection. Initially, they dedicated their definitions and background information to the research ideas and the means of digital safety. The scope and aims of the research were shared with them and then requested to fill out the questionnaire forms.

Components of each period of construction in Fig. 2 is indicated in three steps that uncovers role of digital components in each period of construction. These types of technologies are considered to investigate for applying into construction periods to shape fusion center of safety automation aimed digital transformation implementation in construction building projects.

Due to SEI/ASCE 7_02, each type of building, according to hazard categories, can be modeled by data flow to achieve cost-benefit and optimize the process of management safety. This technique can be used for visualizing and analyzing the flow of information through a system. It's a practical tool for optimizing the design process by identifying and streamlining data exchange between engineers, architects, and other stakeholders involved in seismic design. It also will create a framework for integrating safety performance data into BIM software, which could improve efficiency and communication. Therefore, the application of shaping data fusion centers by the standard approach allows us to compare the safety conditions of different design approaches for a particular building type and hazard category.

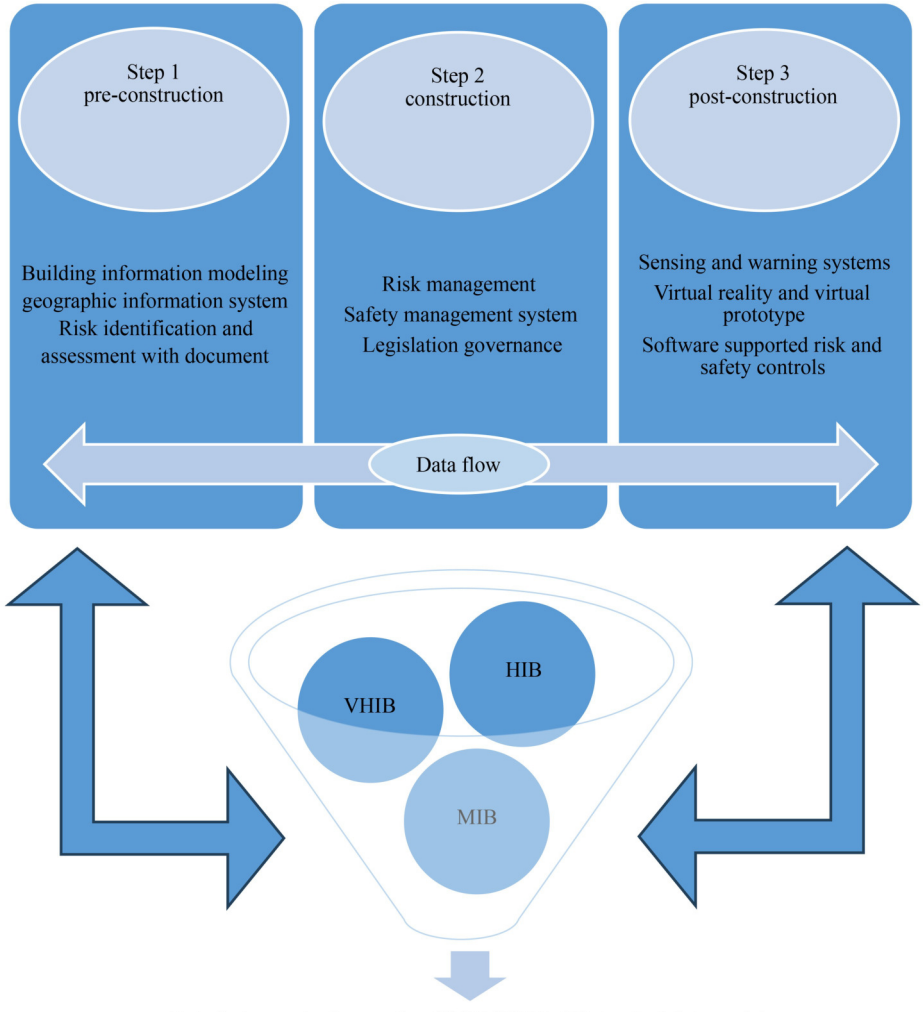


Fig. 2 Extracted model of buildings from American Society of Civil Engineers Standard (SEI/ASCE 7_02) and extracted conceptual model for digital safety in BCM from the study [19].

Due to a standard called SEI/ASCE 7_02, the data have been collected in three categories of buildings' group. 1) Group 1, known as very high importance buildings (VHIB), includes buildings that their post-continuous earthquake operation is of special importance and any discontinuity in this regard leads to an indirect increase in casualties and damages. Hospitals, clinics, fire stations, water supply and power plants, aviation control towers, communication centers including radio and TV, police stations, rescue centers, and, in general, all buildings that are involved in rescue and help operations are in this category. Buildings and structures supporting toxic and hazardous material whose failure may cause widespread important environmental damage in the short and long-term are also considered in this group. 2) Group 2, known as high importance buildings (HIB), includes two types: buildings whose damage results in great loss of life, such as schools, mosques, stadiums, cinemas and movie theaters, assembly halls, departmental stores, terminals or any enclosed area with a capacity greater than 300 people under one ceiling; buildings whose damage results in loss of national heritage. These include museums, libraries, and other places where national documents and valuable items are preserved, as well as industrial buildings and facilities whose failure may result in environmental pollution or widespread fire, such as refineries, fuel storage tanks, and gas supply centers. 3) Group 3, known as moderate importance buildings (MIB), includes all buildings that are within the scope of this Code, except those included in other categories that fall in this group. These include residential, office, and commercial buildings, hotels, multistory parking, warehouses, workshops, industrial buildings, etc.

5 Technique for Order of Preference by Similarity to Ideal Solution results

The results of the decision matrix are indicated in Table 1 for each group of buildings. Value of alternatives and options are evaluated and also their normalized values are represented by their related period's type of building; these types of data indicate in Figs. 3–5.

Options in Table 1 consist of each building's type, and the alternatives are the viewpoint of each respondent's expert in safety and construction. In this Table, each period of construction of every group has been ranked by the values of personal assessment. Elements and options ranking is calculated based on the Consistency Index, which gives the best of the best solution according to the closeness of each item value to the ideal one.

Resultantly, for VHIB, post-construction and PrCPs are in order are consider to prioritize for integration of digitalized safety systems and the values of 0.777 and 0.755 have been proven this theory. In fact, BIM and GIS and also SSRSC, SWS, VR and VP are crucial for merging safety data and developing practical algorithms for modeling new safety solution. Also, pre-construction in HIB as the same with possibility of integration all preferred elements mentioned in Table 1 with final value of 0.905. As Table 1 shows, MIB PrCP is prioritized and besides, SAS and SSRSC may be considered for converging to make a novel paradigm in safety assessment, as well as best of the solutions for making a safe condition of this type of buildings will be, with values of 0.667 and 0.64.

Figure 3 is indicated view point of personal assessment due to collected data which is belonged to VHIB. The details are helped to estimate rank of options in order to find best integrated solution as a digital safety paradigm and to infer technological policies for next generation of digital safety equipment and systems performance. Demographic data and more items about statistics test analysis have noted in 4.1 section of this study. These data are clean version of previous sample, this means that transform data in form of mean values by using compute variables command in SPSS software. Consequently, these types of data used to estimate weighted personal data and final rank of each period of grouped buildings.

Such as Fig. 3, Fig. 4 is indicated of personal assessment data and their weighted values for estimation of final rank of HIB. The weighted is based to entropy method, because degree and efficiency of items of data have placed in heart of the context in this method and this measures value dispersion in decision-making. This satisfying study to achieve more reliable results.

The way of estimation values in Fig. 5 follows the same as previous ones. In this Figure, data are allocated to MIB. For example, according to data, value of 2.33 of the second series of data which is pointed to 20th view point, declares a mean of elements values relate to construction period and actually whole these types of data form decision-based matrix as an input data of TOPSIS algorithm.

As mentioned above data and items of information, it's obvious that PrCP is a critical course of digital safety practices for each building's types as the best option among all while the worse one is CP. This means that digital safety will not be enable and efficient always while buildings operation is in progress, at least an effective side of this insight reveals that digital safety equipment are impactful for planning and surveillance on construction operations. And also, the view point of expert says that cost of handling digital management in construction sites while buildings are constructing is still expensive and they offer to corporate PrCP for handling safety situation.

Figure 6 indicates the rank of the alternatives and following items give more information about each type of buildings.

1) VHIB: PCP period is the best option among all for

Table 1 The value of alternatives and TOPSIS outputs

Personal assessment values	Element	V+	V–	Element Ci	Final Ci
VHIB (Group 1)					
pre-construction	BIM	0.015	0.059	0.797	0.755
	GIS	0.025	0.048	0.658	
	RIA	0.039	0.044	0.53	
construction	RM	0.039	0.037	0.487	0.159
	SMS	0.036	0.045	0.556	
	LG	0.05	0.034	0.405	
post-construction	SWS	0.038	0.042	0.525	0.777
	IES	0.043	0.028	0.394	
	VR and VP	0.031	0.046	0.597	
	SSRSC	0.03	0.052	0.634	
	SAS	0.041	0.035	0.461	
HIB (Group 2)					
pre-construction	BIM	0.014	0.052	0.788	0.905
	GIS	0.026	0.045	0.634	
	RIA	0.02	0.049	0.71	
construction	RM	0.033	0.04	0.548	0.225
	SMS	0.043	0.033	0.434	
	LG	0.046	0.028	0.378	
post-construction	SWS	0.033	0.035	0.515	0.611
	IES	0.04	0.029	0.42	
	VR and VP	0.029	0.045	0.608	
	SSRSC	0.038	0.039	0.506	
	SAS	0.04	0.035	0.467	
MIB (Group 3)					
pre-construction	BIM	0.044	0.046	0.511	0.617
	GIS	0.045	0.044	0.494	
	RIA	0.033	0.048	0.593	
construction	RM	0.046	0.033	0.418	0.445
	SMS	0.059	0.03	0.337	
	LG	0.06	0.026	0.302	
post-construction	SWS	0.058	0.024	0.293	0.424
	IES	0.067	0.018	0.212	
	VR and VP	0.053	0.039	0.424	
	SSRSC	0.031	0.055	0.64	
	SAS	0.031	0.062	0.667	

this type of building with value of 0.777 and in order PrCP with value of 0.755, as the worse option is construction period with its 0.159 value. This means that decision makers first should allocate their safety resources to PCP and PrCP. Insight emphasizes that Fair distribution of construction budget should be lay of this ranked out to practically welcome safety by digital design. this indicates PCP is very importance course to VHIB. Manage and control systems of maintenance in this type of buildings such as HVAC, lighting, safety and security, and also fire alarms need sophisticated applications with integration systems with each other that digital safety management will commit the duty.

2) HIB: in this type of building, priority is allocating the resources to PrCP, due to 0.905 value estimation among other ones with values in order 0.611 and 0.225

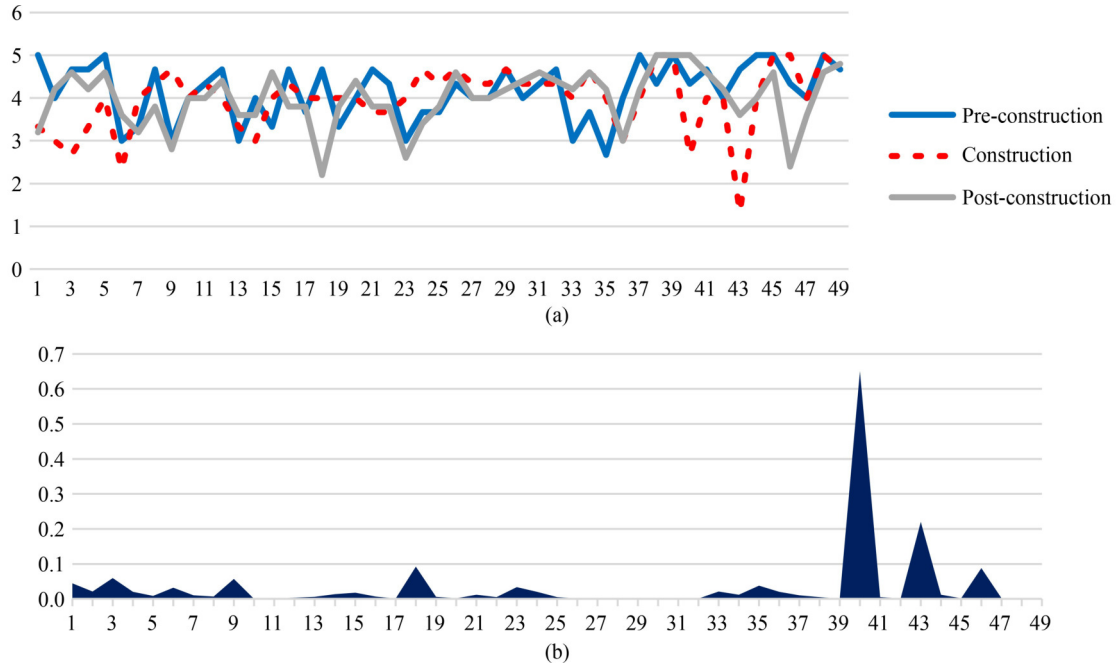


Fig. 3 The data chart of group one belongs to VHIB: (a) collected personal data; (b) weighted personal data.

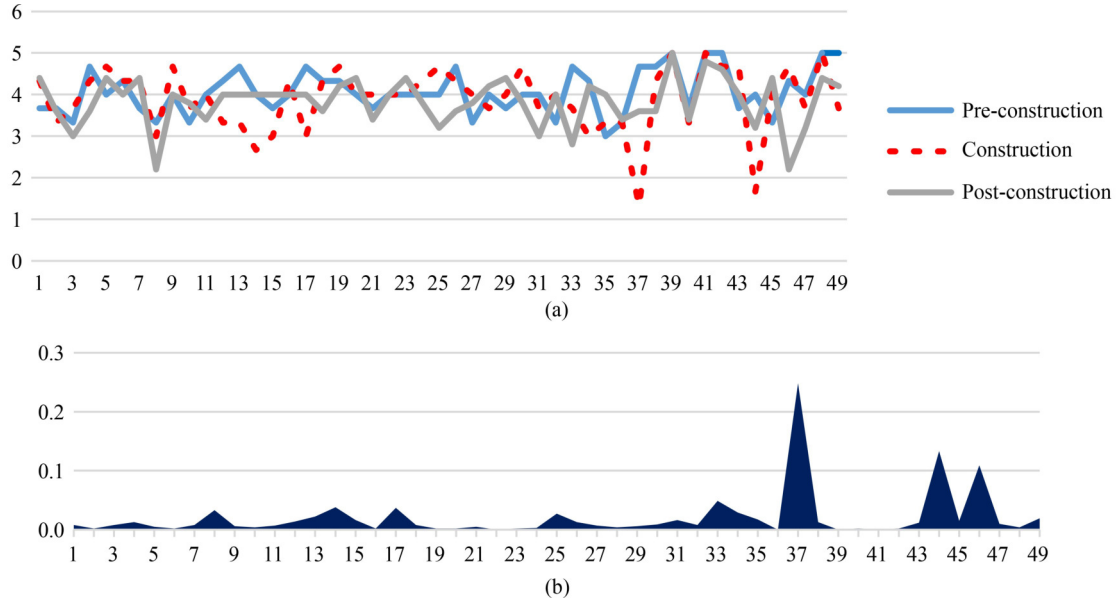


Fig. 4 The data chart of group two belongs to HIB: (a) collected personal data; (b) weighted personal data.

for PCP and CP. Neglect of desirable safety development in earlier phase of projects' HIB lead to unacceptable failures during building's lifespan. Also, construction quality in lifecycle of such buildings increases while safety tact and thoughtfulness apply in suitable situation.

3) MIB: PrCP is top rank for this type of buildings as well as other ones. MIB priorities classify in order of PrCP, CP and PCP in order with their value of 0.617, 0.455, and 0.424.

According to the results mentioned above, PrCP is a primitive priority for enabling digital safety in building construction sites. Then, PCP will be critical to pay more attention on digital safety modeling and as the last order, CP should manipulate digital safety on their operations to reach an acceptable safe condition and also, M.P.B is an important side for prevention against hazards from digital point.

By the use of TOPSIS, decision-makers evaluate and analyze a set of alternatives with multiple attributes to provide desirable alternatives for supporting decision-making. This approach is designed to respond to complex problems, and it is a part of modern decision science [63]. Furthermore, mean values of descriptive statistics are used as input data for TOPSIS, which is mentioned in the

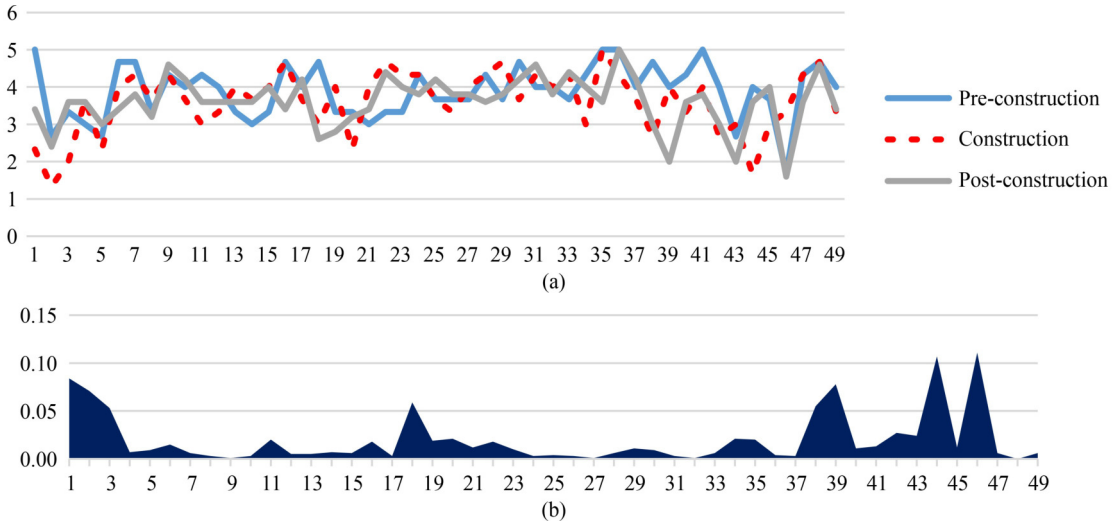


Fig. 5 Data chart of group Three belong to MIB: (a) collected personal data; (b) weighted personal data.

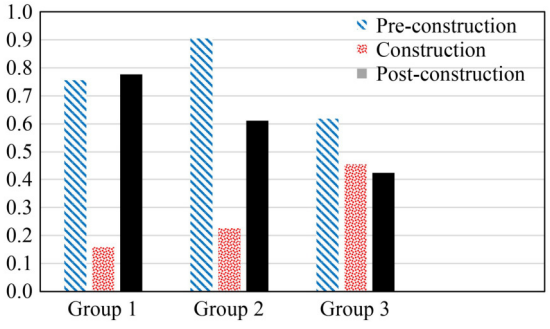


Fig. 6 The rank of the alternatives in three aspects, including PrCP, CP, and PCP.

instruments, which are explained in the following of this research. Its outputs are presented in the part of the research as the data-driven ranking model for each construction periods.

Additionally, decision-makers make their choices based on statistical parameters, and their decisions are based on the best outcome. Their decisions should get away from biases, and they need to make their decisions rationally after researching alternatives and knowing the consequences of their decisions. For this issue, the cumulative sum values for each group are estimated and are shown in Table 2. This will help to estimate percentage values in each group that reflect decision-makers on what period of construction requires an emergency action to enable digital safety.

As above mentioned, the requirement of decision making with approach digital safety reveals. Figure 7 focuses to more emphasizes value by shaping custom combination chart. On the other hand, Fig. 8 shows overall value for each group had mentioned above. As shown in this Figure, all groups need to enable digital safety for control hazardous condition, and its importance is high for all groups. But, group 1 of building types is a priority.

Table 2 Cumulative sum values for each group

Group No.	Pre-construction	Construction	Post construction	Sum of values	Cumulative sums value
1	0.755	0.159	0.777	1.691	1.691
2	0.905	0.225	0.611	1.741	3.432
3	0.617	0.455	0.424	1.496	4.928

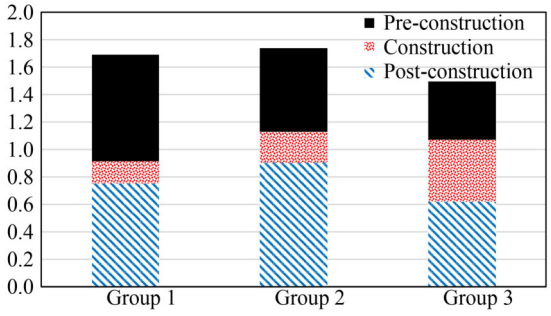


Fig. 7 Achievable TOPSIS values for each group.

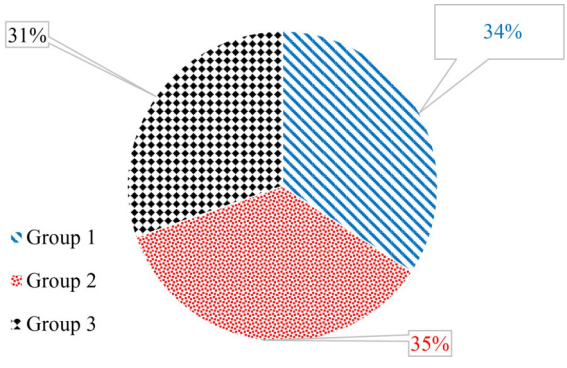


Fig. 8 Overall value for each group.

Group 1 is dedicated to VHIB, and its descriptive data character is demonstrated in Table 3. This Table indicates a critical aspect of data extraction that contributes to the values of examined theories, and it also helps to estimate

Table 3 Group 1 descriptive statistics

Model's elements	Number of cases	Minimum	Maximum	Mean	Std. deviation
BIM	49	2.00	5.00	4.4898	0.79379
GIS	49	2.00	5.00	3.9592	0.81545
RIA	49	1.00	5.00	4.0204	1.07024
RM	49	1.00	5.00	3.8776	1.14805
SMS	49	1.00	5.00	4.0204	0.87773
LG	49	1.00	5.00	4.0612	0.98759
SWS	49	1.00	5.00	3.9796	1.03057
IES	49	1.00	5.00	3.8571	1.00000
VR and VP	49	2.00	5.00	4.0816	0.99659
SSRSC	49	1.00	5.00	4.1224	0.88111
SAS	49	1.00	5.00	3.9184	1.05745

the possibility of integration systems to a system for a safe condition and welcome to safety in construction projects. This will be provided sustainable situation on construction periods by a good governance and process management.

Table 4 indicates descriptive statistics data. These types of data make inferences about the sample population from which the data was drawn. This table contains data for HIB type. This type of data are a powerful tool for summarizing and understanding values of the data-driven model, which is called digital safety for HIB. Indeed, digital safety is the same as of digital safety by design for buildings. This means that technologies will compound each other to refine data and information for serve of safety.

Table 4 Group 2 descriptive statistics

Model's elements	Number of cases	Minimum	Maximum	Mean	Std. deviation
BIM	49	2.00	5.00	4.3061	4.3061
GIS	49	2.00	5.00	3.8776	3.8776
RIA	49	2.00	5.00	4.0204	4.0204
RM	49	2.00	5.00	3.8571	3.8571
SMS	49	1.00	5.00	3.8571	3.8571
LG	49	1.00	5.00	3.9592	3.9592
SWS	49	2.00	5.00	4.0204	4.0204
IES	49	1.00	5.00	3.6327	3.6327
VR and VP	49	2.00	5.00	3.9388	3.9388
SSRSC	49	1.00	5.00	3.6531	3.6531
SAS	49	1.00	5.00	3.8980	3.8980

Table 5 indicates descriptive statistics for group 3, as mentioned above, for other types of buildings. The data set is used to make informed decisions about the data. Additionally, they will help to gain valuable insights for safety improvement and give some tips in safety management activities.

Table 5 Group 3 descriptive statistics

Model's elements	Number of cases	Minimum	Maximum	Mean	Std. deviation
BIM	49	1.00	5.00	1.04002	1.04002
GIS	49	1.00	5.00	0.95253	0.95253
RIA	49	2.00	5.00	1.05342	1.05342
RM	49	2.00	5.00	0.93995	0.93995
SMS	49	1.00	5.00	1.06066	1.06066
LG	49	1.00	5.00	1.13652	1.13652
SWS	49	1.00	5.00	1.06306	1.06306
IES	49	1.00	5.00	1.26740	1.26740
VR and VP	49	1.00	5.00	1.13051	1.13051
SSRSC	49	1.00	5.00	0.91287	0.91287
SAS	49	1.00	5.00	1.09343	1.09343

5.1 Analysis for various period and type of construction

In this study, the Analysis in three periods, including analysis of the pre-construction period, analysis of the construction period, and analysis of the post-construction period, were conducted that in following, they are presented:

5.1.1 Analysis of the pre-construction period

The present period is a critical component of the overall construction process, given its inherent significance in the design phase. There exists a range of empirical evidence that highlights various contentious aspects, including contracts, implementation methods, RM, and predictions. To investigate the potential implementation of digital safety measures during this period, three items were selected for the study: BIM, GIS, and RIA. In summary, Overall, the use of GIS, BIM, and RIA can greatly improve safety in the pre-construction phase of a project. By identifying and mitigating potential risks early in the project lifecycle, construction teams can work more safely and efficiently, ultimately leading to a successful project outcome. Likewise, in the pre-construction phase of a project, several important factors need to be considered to ensure safety during the construction process. GIS and BIM are two technologies that can help identify and assess risks associated with a construction project.

Moreover, RIA plays a crucial role in predicting potential obstacles under uncertain conditions. In uncertain conditions, predicting obstacles can be challenging due to the lack of clear information. However, with careful analysis and contingency planning, potential obstacles can be anticipated and mitigated to ensure successful outcomes. In this regard, a comprehensive questionnaire was developed to address the representative model of the construction building

during uncertain times. Based on the paradigm's outcome, three questions were designed to measure the potential of digital safety modeling during such periods. The RIA framework, coupled with effective modeling techniques, can aid in identifying potential obstacles and mitigating them for successful project outcomes.

5.1.2 Analysis of the construction period

The present study posits that the earlier construction period, encompassing RM, SMS, and LG, is characterized by a notable incidence of prevented accidents or incidents, as well as an emphasis on workers' occupational safety and health policy in order to ensure safe work completion and other safety objectives. The three aforementioned factors serve as representative variables for constructing various related facilities or independent project processes within an overarching construction program. Drawing from the seminal works of Rezaie and Rajabalinejad [84], the current research endeavors to develop a three-item questionnaire survey aimed at assessing the impact of implementing these factors in construction projects based in Tehran, with the aim of validating the theoretical framework underpinning digital safety in construction.

In summary, the relationships between SMS, RM, and LG are crucial during the construction period. SMS helps to minimize potential risks and hazards by implementing safety policies, procedures, and practices, while RM involves identifying and assessing potential risks, developing strategies to mitigate those risks, and monitoring the effectiveness of RM plans. Compliance with LG ensures that construction companies meet the required regulations and legal obligations in the construction industry, thus protecting workers, the public, and the environment during the construction process. Together, these systems work to enhance the overall safety performance of construction projects and reduce accidents and incidents during this critical phase of a project.

5.1.3 Analysis of the post-construction period

PCP is a crucial phase in building construction that ensures the building is safe for occupation and meets all regulatory requirements and standards. It involves a range of activities, including testing and commissioning, issuing occupancy permits, and carrying out repairs or upgrades to the building as necessary, and it refers to the maintenance process pf building. Moreover, quality control is crucial for ensuring that a building is constructed to meet the required specifications. It is a comprehensive process that involves monitoring and testing various aspects of the construction, from materials to workmanship. By implementing effective quality control measures, construction companies can ensure that their projects are safe, high-quality, and meet the expectations of all stakeholders involved.

Therefore, the construction of a building requires strict quality control to ensure that the project is built according to specifications. Additionally, after the completion of the construction phase, maintenance becomes necessary to monitor and identify potential risks. This is where various technological tools such as SWS, IES, VR, VP, SSRSC, and SAS come into play to help prevent risky situations for buildings.

A questionnaire survey was conducted to assess the PCP in terms of safety measures. The survey consisted of five items and was developed based on the requirements gathered from the fieldwork related to safety in building construction.

Overall, the use of technology in ensuring safety in building construction and maintenance has become increasingly important and influential in preventing risky situations. These tools not only aid in preventing accidents but also enable efficient and effective monitoring of potential hazards. It is crucial for builders and maintenance personnel to stay up-to-date with the latest technological advancements and implement them accordingly to ensure the safety of all individuals involved in the building process.

5.2 Reliability and validity of the collected data

In the study, a multi-item survey measure was administered to participants during their working hours. Participants were asked to respond to each item using a five-point Likert-type scale [85]. The purpose of this measure was to assess their perceptions related to a specific construct or factor of interest.

To ensure the quality and validity of the survey items, an initial step involved conducting factor analysis. This analysis examined the factor loadings of each item, which indicates the strength of the association between the item and the targeted construct. Items with a factor loading of 0.3 or higher were considered statistically significant [86], while those with lower loadings were excluded from further analysis.

The survey forms did not require any items to be eliminated because the Kaiser–Meyer–Olkin measure for sampling adequacy exceeded the threshold of 0.6, indicating that there was no need to reduce the factors. Ultimately, by using Cronbach's alpha, the internal consistency reliability of each questionnaire form was evaluated [87] and the value of more than 0.60 is acceptable and approved [88]. For instance, three separate sections, consisting of a total of 11 elements, were used to measure different aspects of the digital context. Each section was considered a reliable indicator of the relevant periods and participants' perception of digital performance in safety. The Cronbach's alpha coefficients

for groups 1, 2, and 3 were 0.671, 0.647, and 0.756, respectively, indicating good internal consistency reliability.

This analysis was conducted in two phases. First, the *t*-test was conducted to see significant differences between the hypotheses in the representative questions. Second, we evaluated how these differences can cause the development of a digital safety model in building construction through field study for proof of research dominance. The unique feature of the sample *t*-test is that it can portray the complex structural framework of digital safety theory as a conceptual model. Analyze assumptions to realize the value of acceptance or rejection and what components are valid for the represented model. It lets this research select its mentioned model for taking feedback in the context of integrating the circulated safe information items to see what happened in their prementioned model in a technical manner.

5.3 Rank data-driven model for integrated data result in construction periods

Data-driven models are increasingly being recognized as an indispensable tool in contemporary business environments. These models leverage systematic data collection and analysis to yield valuable insights that can inform informed decision-making. Their essential value lies in their ability to provide accurate predictions, identify trends, and guide decision-making processes with a high degree of certainty.

In the field of construction safety management, data-driven models offer a crucial means of identifying and mitigating potential hazards. By collecting and analyzing data on past incidents, utilizing predictive analytics, and facilitating communication and collaboration among stakeholders, these models can drive continuous improvement in safety outcomes. As such, data-driven models represent a significant opportunity for organizations to improve their safety performance and reduce the risk of accidents or injuries in construction settings.

Figure 9 normally indicates distributed curves for data-driven models arising from the methodology section. It gives an overview of data sets-related means and standard deviation values to emphasize how data are normally distributed. By uncovering data sets and showing their plots, the researchers were going to characterize data uses and give insight into data acceptable probability. This will be a merit while highlighting the roles of data-driven values and their related insights. Because enhanced values of data will guide the way of choosing some practical solutions.

The normal curves for the first group of the data set series indicate that the data set is normally distributed, with mean values of 4.17, 3.99, and 4.01 The standard deviation of 0.603, 0.732, and 0.643 indicates the amount

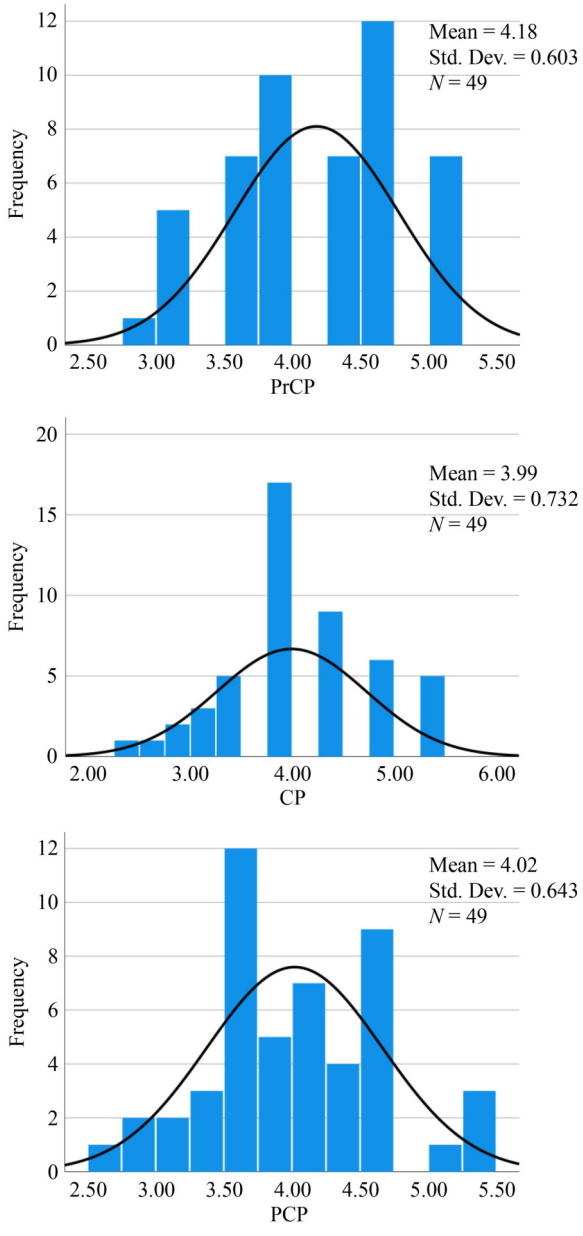


Fig. 9 Distribution analysis of data achievement (Group 1).

of variability or dispersion in the data around the mean value. The values achieved by the data set obtained are worthwhile in ranking. Thus, the value of the integration data-driven model was verified by the sample *t*-test, emphasizing the closeness rate of achieving data to the ideal solution. It is estimated by the TOPSIS method to rank integration data-driven model for high-importance building, which is indicated in Fig. 10.

The importance of uncovering this part by plotting data sets is to improve decision-making by providing a better understanding of the underlying patterns and relationships in the data. The information in Fig. 10 is used to enhance data-driven insights. This allows stakeholders to make better decisions and overcome faced challenges in each period of construction. This is an approach that provides a

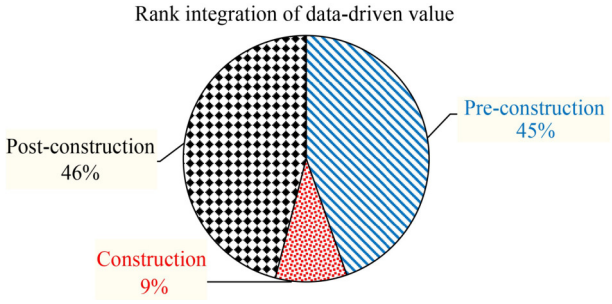


Fig. 10 Maximizing data-driven insights: exploring rank integration for enhanced value (Group 1).

complete and accurate picture of the underlying data for actors in construction sites.

The achievement values of the data set series related to the second group would be centered at 4.09, 3.91, and 3.80, with a spread determined by the standard deviation of 0.49, 0.76, and 0.53 for normally distributed curves, as indicated in Fig. 11. Figure 11 interprets all the data that can be validated for the use of the input ranking system of TOPSIS. It demonstrates how the ideal solution can be selected for a good safety management strategy.

Therefore, as shown in Fig. 12, the value of the integrated data-driven model has been ranked respectively by the value percentage 52% PrCP, 13% CP, and 35% PCP in the TOPSIS system. In this way, the required data insight is given by ranking the achievement data. One of the merits of this process is that it focuses on safety regulations and highlights the importance of enhanced compliance. The PrCP is essential for following the digital safety integration model that is enhanced in this study.

Then, Fig. 13 indicates data for third group. These types of data have related values of 3.92, 3. 61, and 3.60 for mean, and respectively, values of 0.697, 0.760, and 0.638 are standard deviations for distributed analysis of data achievement. The achievable data are normally distributed. It requires ranking by integrated data-driven value to shape the maximum likelihood estimation of each period's construction.

Therefore, Fig. 14 indicates the values of this type of rank mentioned above. Such as two previous examples, its related values ranked at 41% PrCP, 29% PCP, and 30% CP. The best state for improving safety by data-driven values is a PrCP in this group because this period's insights enhance control measures that will be used to mitigate risks of construction and post-construction.

5.3.1 Enhancing digital safety through data-driven management practices

Construction projects are inherently risky endeavors with numerous potential hazards that threaten the safety and well-being of workers. To mitigate these risks, construction firms have traditionally relied on conventional safety

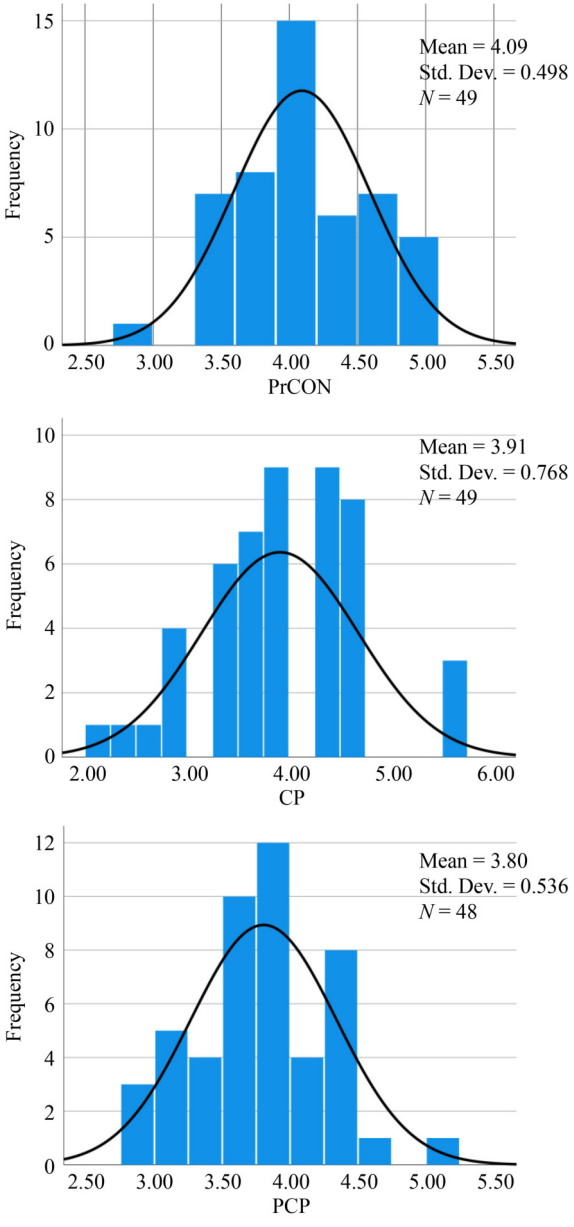


Fig. 11 Distribution analysis of data achievement (Group 2).

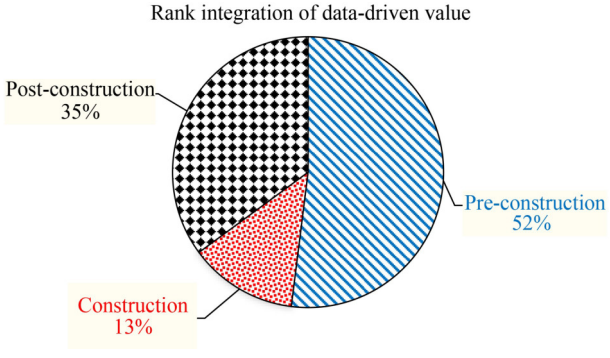


Fig. 12 Maximizing data-driven insights: exploring rank integration for enhanced value (Group 2).

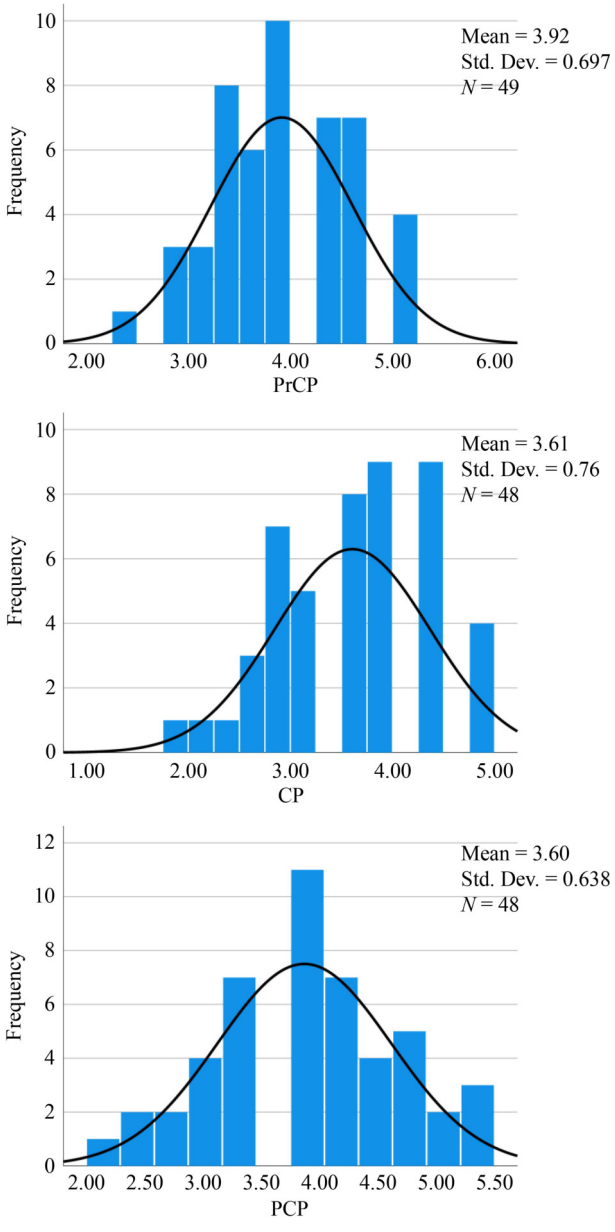


Fig. 13 Distribution analysis of data achievement (Group 3).

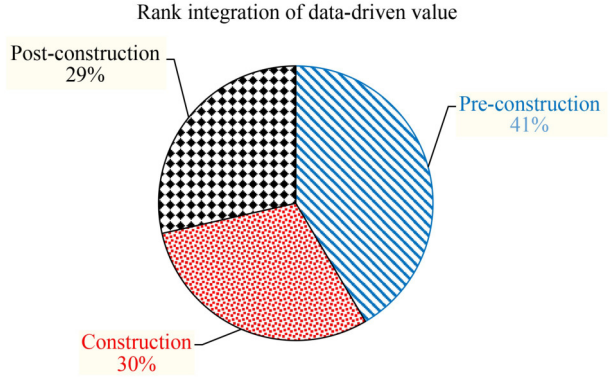


Fig. 14 Maximizing data-driven insights: exploring rank integration for enhanced value (Group 3).

management practices such as training, inspections, and audits. However, advances in digital technology have opened up new opportunities for enhancing safety management through data-driven management practices. By leveraging digital tools to collect and analyze data on various aspects of a project, construction firms can gain valuable insights into potential safety hazards and take proactive steps to address them.

Data-driven management practices enable construction companies to monitor safety performance metrics in real time, identify trends and patterns of behavior, and predict future incidents. Construction companies face numerous safety hazards, which can impact worker safety and increase project costs. To address these risks, firms have traditionally relied on conventional safety management practices such as training and inspections. However, advances in digital technology now provide new opportunities for enhancing safety management through data-driven practices. By leveraging wearable sensors, drones, and other digital tools to collect and analyze data on various aspects of a project, construction firms can gain valuable insights into potential safety hazards and take proactive steps to address them. This approach can lead to improved worker safety, better decision-making, increased efficiency, and cost savings for construction companies.

Finally, the produced values of data-driven management practices offer significant advantages over traditional safety management practices in the construction industry. The use of digital tools to monitor safety performance metrics in real-time, predict potential incidents, and optimize processes can lead to safer work environments, improved project timelines, and lower costs. As such, companies that embrace data-driven safety management practices can create a competitive advantage by reducing risk, improving productivity, and enhancing overall project outcomes.

6 Discussion

The previous study [19] indicated that a DSS as a model is benefit to increase safety in HIB with its conceptual model. This study has tested that approach to other types of buildings and critically interpreted achievable values of data-driven considered approaches. This may shape connectivity in safety for construction management in each period. In this way, it's consumed that an automated approach exposes safety measures for monitoring risks.

This study considers digital safety by a design in construction management that will enabled safety procedures in lifecycle of the building's construction. Table 6 indicates values that refer to digital safety by design model and limitations of acceptance regions for the hypothesis. The data has been validated by the mentioned approach. Automation of digital safety by

Table 6 One sample *t*-test results (digital safety by design)

Personal assessment of values	Element	2-tailed Sig.	*T*-value	*T*-value@95% CI	Using digital safety strategy
VHIB (Group 1)					
pre-construction	BIM	1.634E–17	13.138	1.684	OK
	GIS	9.785E–11	8.234	1.684	
	RIA	2.325E–08	6.674	1.684	
construction	RM	2.000E–06	5.351	1.684	OK
	SMS	1.364E–10	8.138	1.684	
	LG	1.171E–09	7.522	1.684	
post-construction	SWS	2.498E–08	6.654	1.684	OK
	IES	2.507E–07	6.000	1.684	
	VR and VP	8.987E–10	7.597	1.684	
	SSRSC	1.897E–07	8.917	1.684	
	SAS	1.897E–07	6.079	1.684	
HIB (Group 2)					
pre-construction	BIM	2.236E–15	11.487	1.684	OK
	GIS	8.557E–10	7.611	1.684	OK
	RIA	3.385E–09	7.220	1.684	
construction	RM	9.797E–08	6.267	1.684	OK
	SMS	2.507E–07	6.000	1.684	
	LG	7.700E–08	6.335	1.684	
post-construction	SWS	2.613E–11	8.617	1.684	OK
	IES	5.000E–05	4.457	1.684	
	VR and VP	2.487E–09	7.308	1.684	
	SSRSC	1.180E–04	4.191	1.684	
	SAS	1.594E–07	11.487	1.684	
MIB (Group 3)					
pre-construction	BIM	7.215E–09	7.005	1.684	OK
	GIS	2.000E–06	5.399	1.684	
	RIA	4.533E–07	5.831	1.684	
construction	RM	5.000E–06	5.167	1.684	OK
	SMS	6.806E–03	2.828	1.684	
	LG	6.000E–05	4.399	1.684	
post-construction	SWS	1.537E–03	3.360	1.684	OK
	IES	6.131E–02	1.916	1.684	
	VR and VP	7.000E–06	5.055	1.684	
	SSRSC	2.000E–06	5.477	1.684	
	SAS	1.860E–04	4.050	1.684	

design in construction projects involves integrating automated safety measures and technologies into the planning, design, and execution phases of construction to enhance safety, mitigate risks, and improve overall project outcomes. Here's an overview of how automation can shape digital safety in construction projects along with critical points to consider.

1) RIA: automation can assist in identifying potential safety risks in construction projects through tools such as computer-aided design (CAD), BIM, and VR simulations. These technologies enable stakeholders to visualize and evaluate safety hazards before physical construction begins.

Critical Point: it is crucial to ensure that the input data and models used for RIA are accurate, up-to-date, and reflect the actual on-site conditions.

2) Safety Planning and Protocols: automation can streamline the development of safety plans and protocols by generating standardized templates, checklists, and guidelines. Automated systems can help in ensuring compliance with safety regulations and industry best practices.

Critical Point: while automation aids in creating safety plans, human expertise is still necessary to validate and customize these plans based on specific project requirements and contextual factors.

3) Real-time Monitoring and Surveillance: automation enables real-time monitoring of construction sites using various technologies like sensors, cameras, and drones. These systems can detect unsafe conditions, unauthorized access, or equipment failures promptly. Automated surveillance can also identify non-compliance with safety protocols.

Critical Point: automated monitoring systems should be regularly calibrated, properly maintained, and have redundancy measures in place to minimize false alarms or missed safety incidents.

4) IoT and Wearable Technologies: IoT devices and wearables can automate safety-related tasks in construction, such as tracking worker locations, detecting falls, monitoring exposure to hazardous substances, or providing real-time safety alerts. These technologies enhance worker safety and enable quick response to emergencies.

Critical Point: privacy concerns and data security should be addressed when implementing IoT and wearable technologies to ensure sensitive information is adequately protected.

5) Training and Education: automation can support safety training and education by providing interactive e-learning modules, virtual simulations, and augmented reality (AR) tools. These technologies enhance learning effectiveness and retention of safety procedures.

Critical Point: automation should not replace hands-on training and the personalized guidance of experienced personnel. It should complement existing safety training programs.

6) Data Analytics and Predictive Safety: automation enables the collection and analysis of vast amounts of data related to safety incidents, near-misses, and worker behavior. By leveraging ML algorithms, predictive analytics can identify patterns, foresee potential risks, and proactively suggest safety improvements.

Critical Point: Accurate and reliable data collection is vital for effective data analytics. Data privacy and security must be prioritized, and proper anonymization and encryption techniques should be employed.

7) Collaboration and Communication: Automation facilitates seamless communication and collaboration among project stakeholders, enabling quick sharing of safety-related information, incident reports, and updates. Cloud-based platforms, project management tools, and mobile applications foster real-time communication and coordination.

Critical Point: Adequate training and user-friendly interfaces are essential to ensure all stakeholders can effectively utilize collaborative platforms and communicate safety concerns efficiently.

It's important to note that while automation enhances digital safety in construction projects, it should not undermine the significance of human oversight, expertise, and decision-making. Automation should be viewed as a tool that supports and augments human efforts to create safer construction environments.

Additionally, regulatory compliance, stakeholder buy-in, and addressing potential resistance to change are critical factors to consider when shaping automation of digital safety by design in construction projects. Successful implementation requires a comprehensive understanding of project requirements, collaboration among stakeholders, and ongoing evaluation and refinement of automated systems.

Figure 15 is the best way for sharing general insights of this study through digital safety vector. This Figure is indicated conceptual points of digital safety vector. Risky cross point is an initial step of being close to an accident or an incident due its probability and severity. Accident or incident point is defined as time while an accident or an incident occurs. In an uncontrolled situation, accessible to safety items if information is too difficult while it can help to remain in acceptable risk area for controlling risk rate. In contrast, for being in a controlled situation, it's necessary to aware of situation by using data flow and analysis what scenario will happened. Therefore; for avoiding accident or incident point, three principles should be followed. 1) Trying to collecting more safety information for analyzing purposes. 2) Process of management should be integrated into digital safety maturity path for purpose of being in acceptable risk area. 3) Uses data-driven solutions to gain access for more information.

According to Table 6 information, the safety process/action/consequence of items are mentioned in below list and also, it's crucial that using safety strategy is verified for all items of the study. This approach is integrated with ASCE Standard-SEI/ASCE 7-02 and building construction lifecycle process management to make a practical paradigm of digital safety system for all types of buildings.

Pre-construction: 1) BIM should be incorporated into the design process; 2) GIS must be used in selection of structural placement of all type of buildings, especially in low environmental risk impact structures; 3) documentation and risk identification and assessment have to be used as a technical report of pre-construction period.

Construction: 1) risk management is an inseparable part

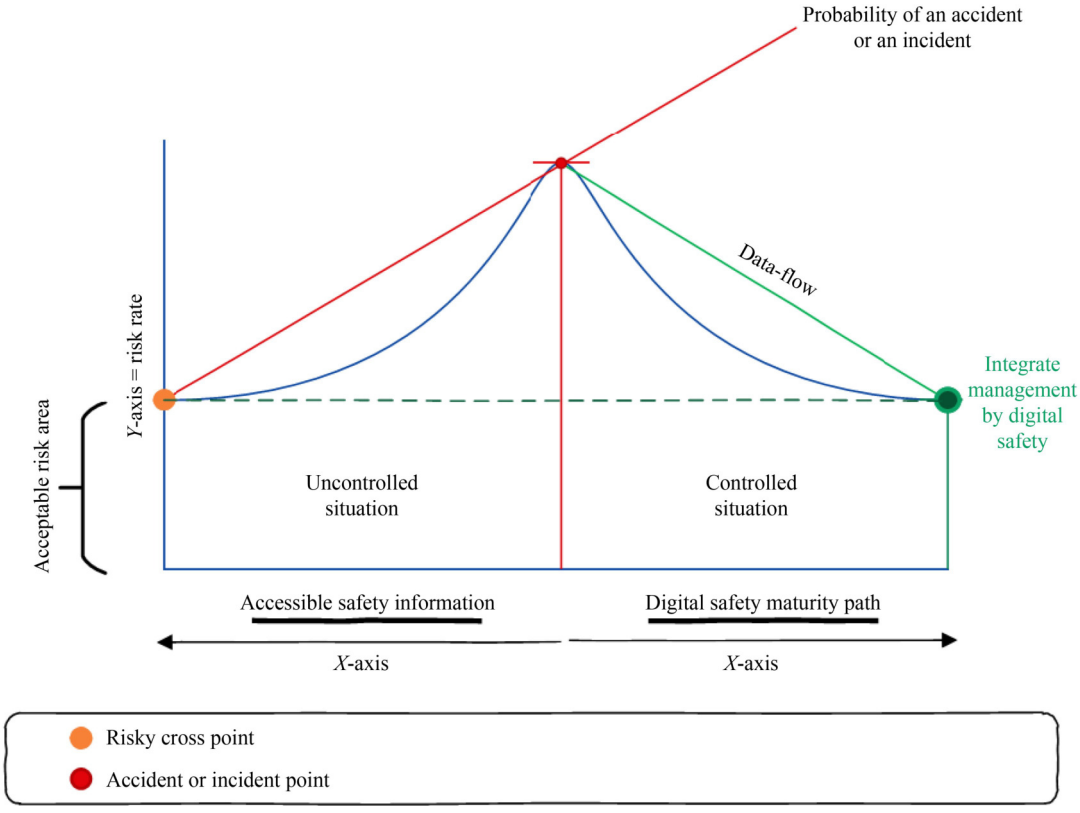


Fig. 15 Conceptual points of digital safety vector.

of construction process; 2) safety management systems must be used as an integral part of construction process to robustness safety culture; 3) legislative governance is a mechanism to manage safety rules, low, and regulation; so, it's needed a compliance designed system by stakeholders of project.

Post-construction: 1)sensing and warning systems play a significant role in monitoring and risky condition for controlling hazards; 2) internet environment services can link into online data-bases to setting configuration out-right to promote decision-making in crisis-time with remote control for any actions; 3) virtual reality and prototyping have to be used in training domain of construction; 4) software's supported risk and safety controls is defined as a critical measurement to make a condition well for who are in-use of constructed structure and must be inhabit in a safe and secure place; 5) safety audit systems must be used to make models of information and improve decisions for future crisis. This part has potential of giving access to safety.

The other relevant information on the above items is given in Table 7. This table gives equality data variances

Table 7 *t*-test for equality of factor means

Variables	*T*-value			Mean difference			2-tailed Sig.		
	Group 1	Group 2	Group 3	Group 1	Group 2	Group 3	Group 1	Group 2	Group 3
BIM	13.138	11.487	7.005	1.490	1.306	1.041	< 0.05	< 0.05	< 0.05
GIS	8.234	7.611	5.399	0.959	0.878	0.735	< 0.05	< 0.05	< 0.05
RIA	6.674	7.220	5.831	1.020	1.020	0.878	< 0.05	< 0.05	< 0.05
RM	5.351	6.267	5.167	0.878	0.857	0.694	< 0.05	< 0.05	< 0.05
SMS	8.138	6.000	2.828	1.020	0.857	0.429	< 0.05	< 0.05	< 0.05
LG	7.522	6.335	4.399	1.061	0.959	0.714	< 0.05	< 0.05	< 0.05
SWS	6.654	8.617	3.360	0.980	1.020	0.510	< 0.05	< 0.05	< 0.05
IES	6.000	4.457	1.916	0.857	0.633	0.347	< 0.05	< 0.05	< 0.05
VR and VP	7.597	7.308	5.055	1.082	0.939	0.816	< 0.05	< 0.05	< 0.05
SSRSC	8.917	4.191	5.477	1.122	0.653	0.714	< 0.05	< 0.05	< 0.05
SAS	6.079	6.129	4.050	0.918	0.898	0.633	< 0.05	< 0.05	< 0.05

in each group compared to mean values. It also gives all items of information about parametric tests and critical values of *t*-statistics for indicating statistically significant differences between the means of the groups.

7 Conclusions

The implementation of safety automation in diverse construction processes necessitates the harmonization and integration of various technologies. This convergence serves as a crucial aspect in establishing a fusion center that promotes efficient hazard prevention and timely risk detection. The present research elucidates the essential technological components that need to interact with one another to realize the notion of digital safety. Digital safety essentially denotes the mechanization of certain safety protocols to enhance safety and mitigate hazards.

Through an examination of several practical components within the realm of digital safety, this study provides confirmation for the hypothesis that the implementation of "Digital Safety" policies on construction sites in the Tehran region is necessary. Such policies would serve to align safety protocols with digital safety approaches, thereby effectively managing and mitigating safety incidents on construction sites.

The research model was found to be acceptable by all participants who completed the questionnaire, indicating a positive reception of the model within the sample population in the Tehran municipality region one.

The statistical package for social science was used for performing a *t*-test of raw scores for each factor. The *t*-values, standard deviations, and 2-tailed significances to test the quality of factor means of different variables for the digital safety factors were divided into the three construction periods shown in Table 7. It has been seen from Table 7 that all digital safety variables, namely BIM, GIS, RIA, RM, SMS, LG, SWS, IES, VR, VP, SSRSC, SAS emerge as significant factors for high-important buildings and very high-important buildings consequences are the same. Still, the variables, namely RM and IES, indicate no tendency as significant factors. As the other type of building, moderate-importance buildings emerge, the variables, namely RM, SMS, SWS., and IES, as insignificant factors; the rest are significant.

The *t*-test of raw scores for each factor was performed using the statistical package for social science. The resulting t-values, standard deviations, and 2-tailed significances were used to assess the quality of factor means across different variables related to digital safety factors, which were divided into three construction periods, as shown in Table 7.

All digital safety variables, including BIM, GIS, RIA, RM, SMS, LG, SWS, IES, VR, VP, SSRSC, and SAS, are significant factors for high-important building and very high-important building consequences. However, variables such as RM and IES do not show a significant tendency.

For moderate-importance buildings, the variables RM, SMS, SWS, and IES are insignificant factors, while the remaining variables remain significant.

To determine the acceptance or rejection area of the given hypotheses, the Sample *t*-test was used to compare a sample's mean to the a priori score [89]. It is because of the declaration of the digital roles in the safety of Tehran construction sites about the items in this study. Detailed items of information about the elements tested are shown in Table 6. It also indicates what sort of digital elements the hypothesis accepts or rejects.

By leveraging advanced technologies and digital safety measures, this model can help protect buildings from potential safety risks, construction incidents or accidents, and other building construction threats. It involves the use of various tools, such as mentioned components of the model to ensure that safety management measures are followed. With a technical manner of digital safety model in place, building owners and managers can rest easy knowing their spaces are protected against any potential digital risks. In this way, the designed model should be examined in the next research domains to emphasize its effects and efficiency.

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Competing interests The authors declare that they have no competing interests.

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